

Practice Problems-7

Two waves traveling together along the same line are given by $y_1 = 5 \sin(\omega t + \pi/2)$ and $y_2 = 7 \sin(\omega t + \pi/3)$. Find (a) the resultant amplitude, (b) the initial phase angle of the resultant, and (c) the resultant equation of motion.

Ans. (a) 11.60, (b) 72.4° , (c) $y = 11.60 \sin(\omega t + 72.4^\circ)$

Three simple harmonic motions are given by $y_1 = 2 \sin(\omega t - 30^\circ)$, $y_2 = 5 \sin(\omega t + 30^\circ)$, and $y_3 = 4 \sin(\omega t + 90^\circ)$. If they are added together, find (a) the resultant amplitude, (b) the initial phase angle of the resultant, and (c) the resultant equation of motion.

Compound graphically two waves having wavelengths in the ratio 3:2 and amplitudes 1:2, respectively. Assume that they start in phase.

Compound graphically two waves having wavelengths in the ratio 4:3 and amplitudes in the ratio 2:3, respectively. Assume that they start in phase.